

8.20 For $n=2$ variable nodes connected by a factor node, the messages will just be passed back and forth along a straight line so it's obvious that the factor node will have received the one incoming message before passing it along.

Now let us assume that for $n=m$ variable nodes, connected in some tree configuration, the nodes will have received all incoming messages before passing them along.

For $n=m+1$ variable nodes, we are basically adding 1 leaf node to the configuration above, via an existing or new factor node. When the messages are passed from leaf nodes to the root node, this node will have the default outgoing message 1.

Since the message passing is starting from leaf nodes, it will be the first to send a message to the factor node it is connected. After receiving the message from the new leaf, the factor node's remaining required incoming messages come from the original m -variable tree, for which we have already assumed that messages can be ordered so that every node receives all necessary incoming messages before sending its outgoing message.

When the messages are propagated from the root node outwards, this new variable node will only be receiving messages and not passing along anything since it is a leaf. So it will not affect the rest of the tree.